

Scatter plot showing USD per bio-benchmark pass (log) for various LLMs and embeddings. The plot compares LLM (API cost) in red triangles, general embeddings in blue circles, and bio-specialised embeddings in green squares. Bio-specialised embeddings like MedEmbed and bge-base-en-v1.5 are highly cost-effective, while general embeddings like jina-embeddings-v5-text-huge and OpenAI Embedding-8B are more expensive. LLMs are the most expensive, with GPT-4o being the most costly.

Model	USD per bio-benchmark pass (log)	Category
pubmedbert-base-embeddings	~10 ⁻³	Embedding, bio-specialised
bge-base-en-v1.5	~10 ⁻³	Embedding, bio-specialised
MedEmbed-small-v0.1	~10 ⁻³	Embedding, bio-specialised
MedEmbed-base-v0.1	~10 ⁻³	Embedding, bio-specialised
multilingual-e5-large-instruct	~10 ⁻²	Embedding, general
embeddinggemma-300m	~10 ⁻²	Embedding, general
jina-embeddings-v5-text-huge	~10 ⁻¹	Embedding, general
OpenAI Embedding-8B	~10 ⁻¹	Embedding, general
gpt-4o	~10 ¹	LLM (API cost)

The scatter plot displays the relationship between the cost of a bio-benchmark pass (USD) and the BioScore (full suite) for different models. The x-axis is logarithmic, representing USD per bio-benchmark pass (log), with major ticks at 10^{-3} , 10^{-2} , 10^{-1} , and 10^0 . The y-axis represents the BioScore (full suite), ranging from 0.54 to 0.59. A vertical dashed line is positioned at 10^{-3} USD. Three models are highlighted with green squares: MedEmbed-base-v0.1 (top left), bge-base-en-v1.5 (middle left), and pubmedbert-base-embeddings (bottom left). Blue circles represent other models, and red triangles represent the baseline. The data points are as follows:

Model	USD per bio-benchmark pass (log)	BioScore (full suite)
MedEmbed-base-v0.1	10^{-3}	0.592
bge-base-en-v1.5	10^{-3}	0.589
pubmedbert-base-embeddings	10^{-3}	0.552
Other Model 1	10^{-3}	0.579
Other Model 2	10^{-3}	0.572
Other Model 3	6×10^{-3}	0.578
Other Model 4	10^{-2}	0.590
Baseline 1	10^0	0.565
Baseline 2	2×10^0	0.538